Ans. (a): In India, rural incomes are generally lower than the urban incomes, because large number of farmers are illiterate and know little about scientific agriculture, Prices of primary products are lower than those of manufactured products and Investment in agriculture has been low when compared to investment in industry.

78. Assertion (A): Though India's national income has gone up several fold since 1947, there has been no marked improvement in the per capita income level.
Reason (R): Sizeable proportion of the population of India is still living below the poverty line.
In the context of the above two statements which one of the following is correct ?
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not a correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (d): Sizeable proportion of the population of India is still living below the poverty line.
Hence option (d) is correct answer.

79. Which one of the following sets of commodities are exported to India by arid and semiarid countries in the Middle East ?
(a) Raw wool and carpets
(b) Fruits and palm oil
(c) Precious stones and pearls
(d) Perfume and coffee

Ans. (b): Fruits and palm oil are exported to India by arid and semi arid countries of Middle East.

80. Consider the following items imported by India :
I. Capital goods
II. Petroleum
III. Pearls and precious stones
IV. Chemicals
V. Iron and Steel
The correct sequence of the decreasing order of these items (as per 94-95 figures), in terms of value, is
(a) I, II, III, IV, V
(b) I, II, IV, III, V
(c) II, I, III, IV, V
(d) II, I, IV, V, III

Ans. (b): According to 1994&95 data that items India imported which decreasing order is - I, II, IV, III, V

81. The table given below depicts the composition of India's exports between 199293 and 199495 :

Items	Years		
	1992-93	1993-94	1994-95
	(Percentage to total)		
Agriculture & allied products	16.9	18.0	15.9
Ores A Minerals	4.0	4.0	3.7
Manufactured foods	75.5	75.6	78.0
Petroleum Products	2.6	1.8	1.9

The changing composition of the export trade is indicative of structural transformation of Indian economy in favour of modernisation. The best indicator of the trend is the
(a) relative shape of petroleum products in exports
(b) decline in the share of agricultural products in exports
(c) constant share of ores and minerals in exports
(d) increase in the share of manufactured products in exports

Ans. (d): The changing composition of the export trade is indicative of structural transformation of Indian economy in favour of modernisation. The best indicator of the trend is the increase in the share of manufactured products in exports.

PHYSICS

82. Consider the following statements: A person in a spaceship located half way between the earth and the sun will notice that the:
1. Sky is jet black
2. Stars do not twinkle
3. Temperature outside the spaceship is much higher than that on the surface of earth
(a) 3 alone is correct (b) 1 and 2 are correct
(c) 1 and 3 are correct (d) 1, 2 and 3 are correct

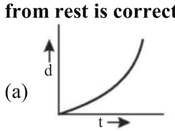
Ans. (d): There is a presence of atmosphere at earth's surface which consists of thick and moving layers of air. The dust particles, particulates, smog, water vapour and smoke are also present in air. Stars twinkle when we see them from the Earth's surface because we are viewing them through thick layer of turbulent (moving) air in the Earth's atmosphere. As their light travels through the many layers of the Earth's atmosphere, the light of the star is bent (reflected) many times in random directions (light is bent when it hits a change in density-like a pocket of cold air or hot air). This random reflection results in the star as its twinkling but for a person in spaceship no such refraction are possible as there is vaccume in space. Thus, star will not twinkle.
Similarly, sky will appear blue to us due to Rayleigh scattering which is again due to presence of atmosphere on earth's surface. The shorter wavelength light is absorbed by the gas molecules of atmosphere. The absorbed blue light is then radiated in different directions. It gets scattered all around the sky. Some of this scattered blue light reaches you. Since, you see, the sky looks blue. Whereas no atmosphere is there in space thus, no absorbing and scattering is possible that's why for a person in spaceship sky appears black.
Temperature outside the spaceship is higher in comparison to earth's surface because of being nearer to sun.

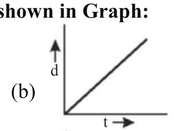
83. When a mirror is rotated by an angle θ, the reflected way will rotate by
(a) 0^0 (b) θ/2
(c) θ (d) 2θ

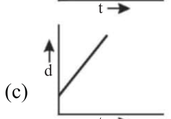
Ans. (d): Consider a plane and a fixed incident ray of light. Before the mirror has rotated, the angle of incidence is θ as is the angle of reflection. If the mirror is rotated through an angle φ the normal is rotated by an angle f and thus the angle of incidence increases to θ + φ. Therefore, the angle of reflection must also increase by φ to θ + φ. The difference between the final angle of reflection and the initial angle of reflection is 2φ. Thus for a fixed incident ray the angle of the reflection is twice the angle through which the mirror has rotated.

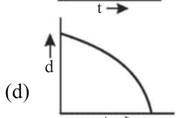
84. Which one of the following layers of atmosphere is responsible for the deflection of radio waves.
(a) Troposphere (b) Stratosphere
(c) Mesosphere (d) Ionosphere

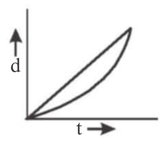
Ans. (d): Ionosphere is the top layer of atmosphere. Radio waves are deflected in the ionosphere of the atmosphere ionosphere is composed of DE and F layers. D layers is good at absorbing AM radio waves.

85. The variation of displacement d with time t in the case of a particle falling freely under gravity from rest is correctly shown in Graph:
(a) 

(b) 

(c) 

(d) 

Ans. (a): As
$\alpha = UT + \frac{1}{2}gt^2$
U = 0 (given)
$\alpha = \frac{1}{2}gt^2$
$\alpha \propto +2$


86. Optical Fibre works on the principle of:
(a) total internal reflection
(b) interference
(c) scattering
(d) interference

Ans. (a): An optical fibre is a thin, flexible, transparent fibre that acts as waveguide or "light pipe" to transmit light between the two ends of the fibre. An optical fibre transmits light along its axis by the process of total internal reflection. When light traveling in a dense medium hits a boundary at an angle larger than the "critical angle" for the boundary, the light will be completely reflected. This effect is used in optical fibres to confine light in the core.

87. Given below are to statements, one labeled as assertion (A) and other labeled as reason (R)
Assertion (A): Transformer is useful for stepping up or stepping down voltages.
Reason (R): Transformer is a device used in D.C. circuits. In the context of above two statements, which one of the following is correct?
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (c): A transformation is a device that transfers electrical energy from one circuit to another through inductively coupled conductor-the transformer's coils. A varying current in the first or primary winding creates a varying magnetic flux in the transformer's core and thus a varying magnetic field through the secondary winding. This varying magnetic field induces a voltage in the secondary winding. By appropriate selection of the ratio of turns, a transformer thus allows an alternating current (AC) or voltage to be "stepped up" by making N_s greater than N_p or "stepped down" by making N_s less than N_p.
Here N_p represents number of turns in primary winding. N_s represents the number of turns in secondary winding. As it is clear from principle involved that transformer can be used to step up and and step down A.C. voltage only. This device and not be used for D.C. voltage as its working is based on the change of magnetic flux with varying current.